Vectors

Q1. Introduction to vectors. Let $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = -\mathbf{i} + \mathbf{j}$.

On the same set of axes draw the vectors:

(i) $\mathbf{a}$ (ii) $\mathbf{b}$ (iii) $-\mathbf{a}$ (iv) $2\mathbf{b}$ (v) $\mathbf{a} + \mathbf{b}$ (vi) $\mathbf{b} - 3\mathbf{a}$

Q2. Vector Arithmetic. Let $\mathbf{u} = 3\mathbf{i} + \mathbf{j}$, $\mathbf{v} = 2\mathbf{i} + \mathbf{k}$, $\mathbf{w} = -\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ be vectors in $\mathbb{R}^3$. Find

(a) $2\mathbf{u}$ (b) $-3\mathbf{u}$ (c) $\frac{1}{2}\mathbf{w}$

(d) $\mathbf{u} + \mathbf{v}$ (e) $\mathbf{w} - \mathbf{v}$ (f) $\mathbf{u} - 5\mathbf{v} + 2\mathbf{w}$

Position and Length

Q3. Position and Length. For each of the following pairs of points find:

(i) the position vector in the direction from the first point to the second point

(ii) the length of this vector.

(a) $(0, 2)$ $(4, -5)$ (b) $(2, 3)$ $(5, 4)$ (c) $(4, -5)$ $(0, 2)$

(d) $(5, 4)$ $(2, 3)$ (e) $(3, 7)$ $(5, 7)$ (f) $(7, -3)$ $(3, -3)$

Q4. Unit Length, Unit Vector. Calculate unit vectors in the direction of each of the position vectors you have found in the previous question.

Q5. Let's Go Backwards. For each pair of vectors above, find the position vector in the direction from the second point to the first point.

Q6. More Vector Lengths. Let $\mathbf{u} = 3\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ and let $\mathbf{v}$, the vector from the point $(2, 1, -1)$ to $(3, 2, -1)$, be vectors in $\mathbb{R}^3$. Find

(a) $\|\mathbf{u}\|$ (b) $\|\mathbf{v}\|$ (c) $\hat{\mathbf{u}}$

(d) $\hat{\mathbf{v}}$ (e) $2\mathbf{u} - \mathbf{v}$ (f) $\|2\mathbf{u} - \mathbf{v}\|$

(g) A unit vector parallel and in the same direction as $2\mathbf{u} - \mathbf{v}$

Scalar Product and Angles

Q7. Scalar product. Computre $\mathbf{u} \cdot \mathbf{v}$ for each of the following pairs of vectors:

(a) $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ (b) $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$

(c) $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} - \mathbf{j} - \mathbf{k}$ (d) $\mathbf{u} = \mathbf{i} + \mathbf{j}$ and $\mathbf{v} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$

(e) $\mathbf{u} = \mathbf{j} + \mathbf{k}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j}$ (f) $\mathbf{u} = \mathbf{i} + \mathbf{j}$ and $\mathbf{v} = \mathbf{k}$

Q8. Angles. Find the angle between each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ above.

Q9. Scalar Product and Angles If $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 3\mathbf{j} + \mathbf{k}$, find:

(a) $\mathbf{a} \cdot \mathbf{b}$

(b) the angle between $\mathbf{a}$ and $\mathbf{b}$

Q10. Perpendicular. Find a constant a in each case below, such that the pairs of vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

(a) $\mathbf{u} = 3\mathbf{i} + a\mathbf{j}$ and $\mathbf{v} = 2\mathbf{i} + 6\mathbf{j}$

(b) $\mathbf{u} = a\mathbf{i} - 3a\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 4\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

Q11. Parallel and Perpendicular. If $\mathbf{u} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ which of the following are either parallel or perpendicular to $\mathbf{u}$?

(a) $-3\mathbf{i} - 4\mathbf{j} - 5\mathbf{k}$

(b) $5\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$

(c) $-3\mathbf{i}$

(d) $-5\mathbf{i} + 3\mathbf{k}$

(e) $-3\mathbf{j} + 5\mathbf{k}$

Q12. Vector Angles. Let $\mathbf{a} = (2, -1, 6)$ and $\mathbf{b} = (1, -1, -1)$.

(a) Show that $\mathbf{a} + \mathbf{b}$ is perpendicular to $\mathbf{b}$.

(b) Find the angle between the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$.

Vector Projections

Q13. Projections in $\mathbb{R}^2$. Let $\mathbf{a} = (-3, 4)$, $\mathbf{b} = (0, 2)$ and $\mathbf{c} = (3, -1)$.

(a) Find $\mathbf{b}_{\parallel} = \text{proj}_{\mathbf{a}}(\mathbf{b})$, the vector projection of $\mathbf{b}$ in the direction parallel to $\mathbf{a}$.

(b) Find $\mathbf{c}_{\parallel} = \text{proj}_{\mathbf{a}}(\mathbf{c})$, the vector projection of $\mathbf{c}$ in the direction parallel to $\mathbf{a}$.

(c) On the same set of x-y axes draw the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\text{proj}_{\mathbf{a}}(\mathbf{b})$ and $\text{proj}_{\mathbf{a}}(\mathbf{c})$.

Q14. Projection in $\mathbb{R}^3$ For each of the pairs of vectors in $\mathbb{R}^3$, find (i) the vector projection of $\mathbf{v}$ in the direction parallel to $\mathbf{u}$, (ii) the vector projection of $\mathbf{v}$ in the direction perpendicular to $\mathbf{u}$

(a) $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 4\mathbf{i} + \mathbf{j}$

(b) $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

(c) $\mathbf{u} = 2\mathbf{i} - 3\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$

(d) $\mathbf{u} = 5\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + 2\mathbf{k}$

(e) $\mathbf{u} = \mathbf{j} - 2\mathbf{k}$ and $\mathbf{v} = \mathbf{i} - \mathbf{k}$

(f) $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{v} = \mathbf{j}$

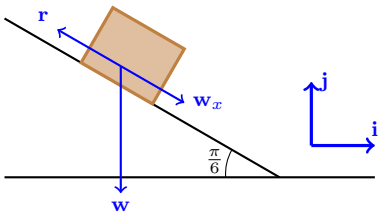
Q15. Points and lines. Let L be the line passing through the points $A = (1, 1, 2)$ and $B = (3, 5, 6)$.

(a) Use a vector projection to find the closest point P on L to the point $C = (2, 1, 1)$.

(b) Find the distance between P and C.

Q16. Gravity. Assume that a box with mass $m = 50$ kgs is on a ramp as shown below. The magnitude of the weight force, $\mathbf{w}$, on the box is mg Newtons/sec in a direction vertically downwards, where $g = 9.8$ m/sec^2. In order to stop the box sliding a force of $\mathbf{r} = -\mathbf{w}_x$ is

applied in the direction shown where $\mathbf{w}_x$ is the projection of $\mathbf{w}$ in the direction parallel to the ramp. Find the following:



(a) The vector $\mathbf{w}$ representing the gravitational force.
(b) A vector $\mathbf{u}$ representing the ramp direction.
(c) The projection $\mathbf{w}_x = \mathbf{w}_{\|}$ onto $\mathbf{u}$.
(d) The required force vector $\mathbf{r} = -\mathbf{w}_x$.
(e) The magnitude of the required force.

Parametric Curves

Q17. Equation of a Path. Let a particle's position as a function of time be as given below. Find the equation of the path for time $t \geq 0$. Sketch the graph of the path.

(a) $\mathbf{r}(t) = t\mathbf{i} - 2t\mathbf{j}$

(b) $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j}$

(c) $\mathbf{r}(t) = (t-4)\mathbf{i} + 3t^2\mathbf{j}$

(d) $\mathbf{r}(t) = t^3\mathbf{i} + t\mathbf{j}$

(e) $\mathbf{r}(t) = (t+1)\mathbf{i} + (t^2 - 2t)\mathbf{j}$

Q18. Periodic Motion. For each of the following functions describing a particle's position at time t, (i) find the Cartesian equation of the path, (ii) sketch its graph, (iii) state the period of motion of the path.

(a) $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\mathbf{j}$

(b) $\mathbf{r}(t) = 2\cos(2t)\mathbf{i} + 2\sin(2t)\mathbf{j}$

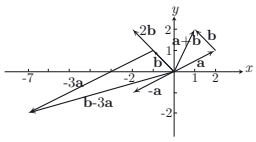
(c) $\mathbf{r}(t) = (2 + \cos(t))\mathbf{i} + (-3 + \sin(t))\mathbf{j}$

Q19. Another Path. Let $\mathbf{r}(t) = (e^t - e^{-t})\mathbf{i} + (e^t + e^{-t})\mathbf{j} = x\mathbf{i} + y\mathbf{j}$ where $t \in \mathbb{R}$.

(i) Find $y^2 - x^2$.

(ii) Find the Cartesian equation of the path and sketch this path.

A1. Introduction to vectors.



A2. Vector Arithmetic.

(a) $6\mathbf{i}+2\mathbf{j}$ (b) $-9\mathbf{i}-3\mathbf{j}$ (c) $-\frac{1}{2}\mathbf{i}-\mathbf{j}+\frac{3}{2}\mathbf{k}$

(d) $5\mathbf{i}+\mathbf{j}+\mathbf{k}$ (e) $-3\mathbf{i}-2\mathbf{j}+2\mathbf{k}$ (f) $-9\mathbf{i}-3\mathbf{j}+\mathbf{k}$

A3. Position and Length.

(a) (i) $4\mathbf{i}-7\mathbf{j}$ (ii) $\sqrt{65}$ (b) (i) $3\mathbf{i}+\mathbf{j}$ (ii) $\sqrt{10}$

(c) (i) $-4\mathbf{i}+7\mathbf{j}$ (ii) $\sqrt{65}$ (d) (i) $-3\mathbf{i}-\mathbf{j}$ (ii) $\sqrt{10}$

(e) (i) $2\mathbf{i}$ (ii) 2 (f) (i) $-4\mathbf{i}$ (ii) 4

A4. Unit Length, Unit Vector.

(a) $\frac{1}{\sqrt{65}}(4\mathbf{i}-7\mathbf{j})$ (b) $\frac{1}{\sqrt{10}}(3\mathbf{i}+\mathbf{j})$ (c) $\frac{1}{\sqrt{65}}(-4\mathbf{i}+7\mathbf{j})$

(d) $\frac{1}{\sqrt{10}}(-3\mathbf{i}-\mathbf{j})$ (e) $\mathbf{i}$ (f) $-\mathbf{i}$

A5. Let's Go Backwards.

(a) $-4\mathbf{i}+7\mathbf{j}$ (b) $-3\mathbf{i}-\mathbf{j}$ (c) $4\mathbf{i}-7\mathbf{j}$

(d) $3\mathbf{i}+\mathbf{j}$ (e) $-2\mathbf{i}$ (f) $4\mathbf{i}$

A6. More Vector Lengths.

(a) $\sqrt{26}$ (b) $\sqrt{2}$ (c) $\frac{3}{\sqrt{26}}\mathbf{i}-\frac{4}{\sqrt{26}}\mathbf{j}+\frac{1}{\sqrt{26}}\mathbf{k}$

(d) $\frac{1}{\sqrt{2}}\mathbf{i}+\frac{1}{\sqrt{2}}\mathbf{j}$ (e) $5\mathbf{i}-9\mathbf{j}+2\mathbf{k}$ (f) $\sqrt{110}$

(g) $\frac{1}{\sqrt{110}}(5\mathbf{i}-9\mathbf{j}+2\mathbf{k})$

A7. Scalar product. (a) 7 (b) 3 (c) 1 (d) 1 (e) 1 (f) 0

A8. Angles. (a) $\arccos\left(\frac{7}{\sqrt{84}}\right)$ (b) $\arccos\left(\frac{3}{\sqrt{84}}\right)$ (c) $\arccos\left(\frac{1}{\sqrt{33}}\right)$
(d) $\arccos\left(\frac{1}{\sqrt{12}}\right)$ (e) $\frac{\pi}{3}$ (f) $\frac{\pi}{2}$

A9. Scalar Product and Angles (a) 4 (b) $\arccos\left(\frac{4}{\sqrt{14}\sqrt{11}}\right)$

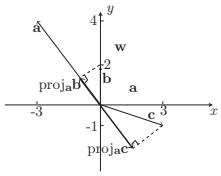
A10. Perpendicular. (a) $a=-1$ (b) $a=-\frac{4}{5}$

A11. Parallel and Perpendicular. (a) is parallel and (d) is perpendicular

A12. Vector Angles. (a) $(\mathbf{a}+\mathbf{b})\cdot\mathbf{b}=0$. (b) $\arccos\left(\frac{38}{\sqrt{38}\sqrt{50}}\right)=\arccos\left(\frac{\sqrt{19}}{5}\right)$

A13. Projections in $\mathbb{R}^2$. (a) $\frac{8}{25}(-3,4)$ (b) $-\frac{13}{25}(-3,4)$

(c)



A14. Projection in $\mathbb{R}^3$

(a) (i) $\mathbf{u}$ (ii) $\mathbf{i}-\mathbf{j}-\mathbf{k}$ (b) (i) $\frac{5}{7}\mathbf{u}$ (ii) $\frac{1}{7}(16\mathbf{i}-4\mathbf{j}-8\mathbf{k})$ (c) (i) $\frac{7}{13}\mathbf{u}$ (ii) $\frac{1}{13}(12\mathbf{i}+13\mathbf{j}+8\mathbf{k})$
(d) (i) $\frac{4}{35}\mathbf{u}$ (ii) $\frac{1}{35}(50\mathbf{i}-4\mathbf{j}+82\mathbf{k})$ (e) (i) $\frac{2}{5}\mathbf{u}$ (ii) $\frac{1}{5}(5\mathbf{i}-2\mathbf{j}-1\mathbf{k})$ (f) (i) $\frac{1}{3}\mathbf{u}$ (ii) $\frac{1}{3}(-\mathbf{i}+2\mathbf{j}-\mathbf{k})$

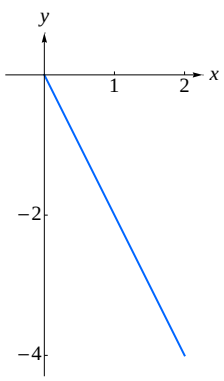
A15. Points and lines. (a) $\frac{1}{9}(8,7,16)$. (b) $\frac{\sqrt{17}}{3}$

A16. Gravity.

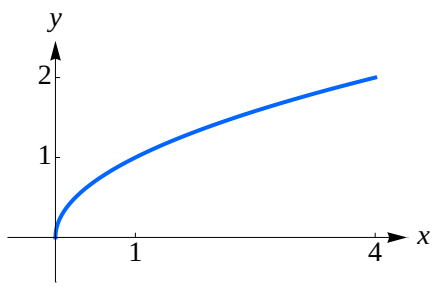
(a) $\mathbf{w}=-490\mathbf{j}$. (b) $\mathbf{u}=-\sqrt{3}\mathbf{i}+\mathbf{j}$. (c) $\mathbf{w}_x=\frac{245\sqrt{3}}{2}\mathbf{i}-\frac{245}{2}\mathbf{j}$. (d) $\mathbf{r}=-\frac{245\sqrt{3}}{2}\mathbf{i}+\frac{245}{2}\mathbf{k}$. (e) $\|\mathbf{r}\|=245$.

A17. Equation of a Path.

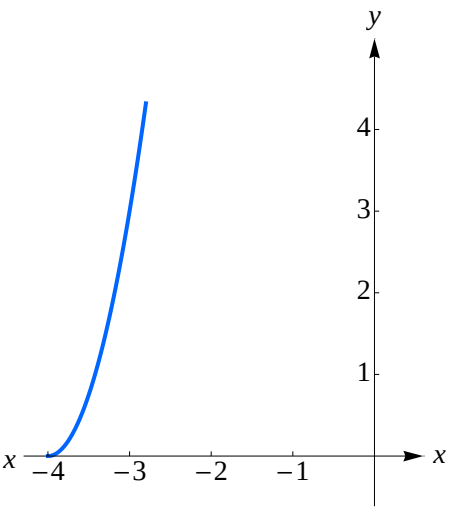
(a) $y=-2x$



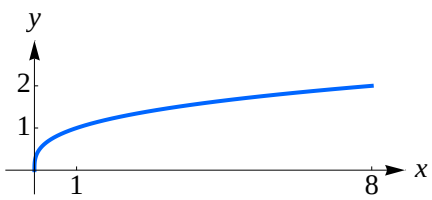
(b) $y=\sqrt{x}$



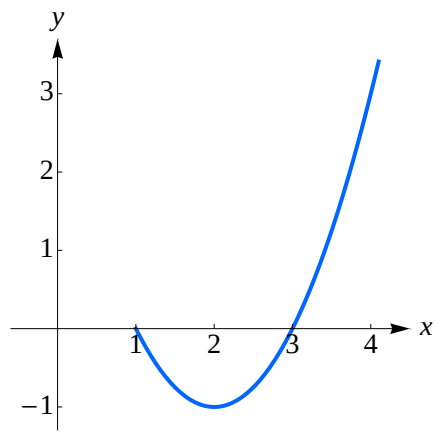
(c) $y=3(x+4)^2$



(d) $y=x^{\frac{1}{3}}$

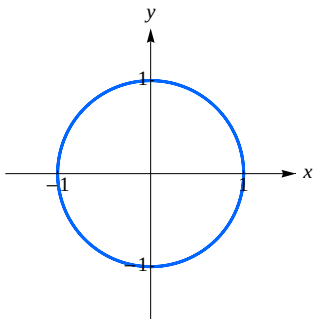


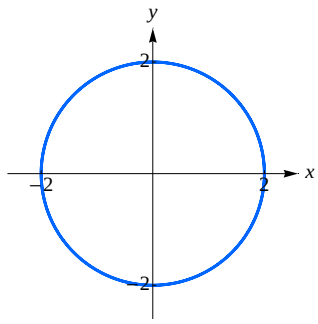
(e) $y=(x-3)(x-1)$

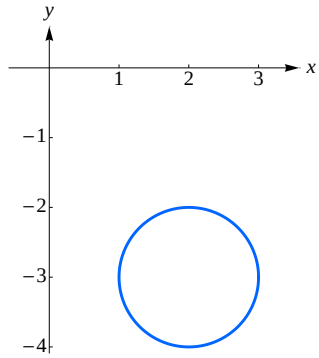


A18. Periodic Motion.

(a) $x^2 + y^2 = 1$, period π (b) $x^2 + y^2 = 4$, period π (c) $(x-2)^2 + (y+3)^2 = 1$, period 2π

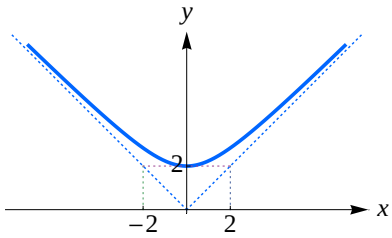






A19. Another Path.

(i) $y^2 - x^2 = 4$ (ii) This is a hyperbola and can be written $y^2/4 - x^2/4 = 1$. Note that $y \geq 0$ so it is only part of the hyperbola. The asymptotes are $y = \pm x$.



Useful Formulae

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1$$

Compound angle formulae

$$\begin{aligned}
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\tan(x+y) &= \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}
\end{aligned}$$

Derivatives of inverse trigonometric functions

$$\begin{aligned}
\arcsin'(x) &= \frac{1}{\sqrt{1-x^2}} \\
\arccos'(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\arctan'(x) &= \frac{1}{1+x^2}
\end{aligned}$$

Antiderivatives from inverse trigonometric functions

$$\begin{aligned}
\int \frac{1}{\sqrt{s^2-x^2}}\,dx &= \arcsin\left(\frac{x}{s}\right) + C \\
\int \frac{-1}{\sqrt{s^2-x^2}}\,dx &= \arccos\left(\frac{x}{s}\right) + C \\
\int \frac{1}{s^2+x^2}\,dx &= \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C
\end{aligned}$$

where s is a *positive* constant, and C is an arbitrary constant of integration.

Complex exponential formulae

$$\begin{aligned}
e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\
\cos(\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \\
\sin(\theta) &= \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)
\end{aligned}$$

Vector projections

- The *vector projection* of $\mathbf{v}$ onto $\mathbf{u}$ is $\mathbf{v}_{\parallel} = (\hat{\mathbf{u}} \cdot \mathbf{v})\hat{\mathbf{u}} = k\mathbf{u}$, where $k \in \mathbb{R}$ is the unique solution of $\mathbf{u}\cdot(\mathbf{v}-k\mathbf{u}) = 0$.
- The *vector component of* $\mathbf{v}$ *perpendicular to* $\mathbf{u}$ is $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$.

Complex roots

The n-th roots of $w = se^{i\phi}$ are $s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\phi+2k\pi))}$ for $k = 0, 1, \ldots, n-1$.

Changes in speed

Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the speed function $\|\mathbf{r}'(t)\|$ is decreasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) < 0$ and increasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) > 0$.